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Sera of Elderly Bullous Pemphigoid Patients with Associated Neurological Diseases Recognize Bullous Pemphigoid Antigens in the Human Brain

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Key Words
Bullous pemphigoid · Neurological disease · Bullous pemphigoid antigen 1 · Skin · Brain · Immunoblotting · Serum

Abstract
Background: Bullous pemphigoid (BP) is an acquired autoimmune dermatosis that is often associated with various neurological diseases (ND) in the elderly. Previously, we reported that BP antigen 1 (BPAG1) of the mouse brain can be recognized by serum samples of BP patients with ND (BP+ND), indicating the possibility that BPAG1 could act as a shared autoantigen of the skin and brain. However, it is not known whether the serum of BP+ND patients could recognize BPAG1 in the human brain. ***Objectives:*** The aim of this study is to determine whether sera of BP+ND patients could recognize BP antigens in human brain tissue. ***Methods:*** Serum samples of BP+ND patients and those of age- and sex-matched BP patients without ND, ND patients without BP, and normal controls were obtained. Serum samples were examined by immunoblotting against protein extracts of human skin and brain. ***Results:*** A 230-kDa protein of human epidermal extract was recognized by 16 of 22 (72.7%) serum samples of elderly BP+ND patients, whereas it was recognized by 50, 0, and 0% of serum samples of BP, ND, and controls, respectively. A 230-kDa protein of human brain extract was recognized by 12 of 22 (54.5%) serum samples of BP+ND patients, whereas it was recognized by 9.1, 9.1, and 4.5% of serum samples of BP, ND, and controls, respectively. A 180- and a 165-kDa protein could also be recognized by a proportion of serum samples of BP and BP+ND patients. ***Conclusion:*** This study demonstrated that both human skin and brain contain immunogenic BPAG1 in BP+ND patients. We speculate that alterations of the central nervous system in the course of pathological changes of elderly ND patients could expose the neural isoforms of BPAG1 or other BP antigens. Autoantibodies against these antigens in the brain may cross-react with antigen in the skin, which may underlie the pathological development of BP.

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Introduction

Bullous pemphigoid (BP) is an autoimmune bullous disorder of the skin that most commonly affects the elderly [1]. BP is caused by immune reactions of autoanti-

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bodies against BP antigens (BPAG1 and BPAG2) in the basal membrane of the skin [1]. Three major forms of BPAG1 exist. The neural isoform of BPAG1-a and the muscle isoform of BPAG1-b are highly expressed in the brain and heart, respectively [2]. The epithelial isoform BPAG1-e is only found in the skin, in which BPAG1-a and BPAG1-b mRNAs are also present, but at lower levels [2]. Because of overlapping expression patterns in the skin and sequence similarities between BPAG1-e and BPAG1-a [1, 3, 4], immune cross-reactions might exist in BP patients with neurological diseases (ND).

The prevalence of BP is elevated in ND patients. In comparison to acute brain damage (such as trauma [1, 4]), neurological disorders (such as dementia, cerebral stroke, and Parkinson's disease) are commonly associated with BP [1, 3–5]. Conversely, there is also a high prevalence of ND in BP patients [3, 5]. The development of ND often precedes the onset of BP. The interval between the diagnosis of ND and BP ranges from months to years [1, 3]. These clinical associations suggest that pathogenic relationships exist between BP and ND.

Pathogenic relationships between BP and ND are not clear. Establishing such relationships is important to understand the development and progression of BP. Previously, we found that serum samples of BP and ND (BP+ND) patients could recognize BPAG1 and BPAG2 in human skin and mouse brain [6], raising the possibility that BPAG1 could act as a shared autoantigen of the skin and brain in humans. In the present study, we explored the possibility that immunological reactions exist between BP autoantibodies and human brain tissue. Immunoblotting demonstrated that a significant proportion of serum samples obtained from BP+ND patients could indeed react with antigens of the human brain, confirming the existence of BP antigens in the brain of elderly ND patients, which could contribute to the development of BP.

Methods

Patient Selection

Four groups of age- and sex-matched patients were included in this study:

Bullous Pemphigoid with Neurological Diseases (BP+ND). Twenty-two BP patients with ND. Among these patients, 12 were also included in a previous study [6]. BP was diagnosed in the Peking Union Medical College Hospital (PUMCH) from 2001 to 2008 according to the following criteria: (1) characteristic clinical presentation of tense cutaneous blisters; (2) histopathological evidence of subepidermal blister formation; (3) linear IgG and/or C3 deposition at the dermal-epidermal junction of the perilesional skin by direct immunofluorescence labeling [7]. NDs were diagnosed by criteria according to the patients' clinical and imaging (MRI or CT scan) examinations, including 8 cases of dementia, 9 cases of cerebral thrombosis, and 5 cases of cerebral hemorrhage.

Bullous Pemphigoid (BP). Twenty-two BP patients without ND. These patients were diagnosed as having BP by criteria described for the BP+ND group in PUMCH (clinical data not shown).

Neurological Diseases (ND). Twenty-two ND patients without BP. These patients were diagnosed with ND by the same criteria as described for the BP+ND group (clinical data not shown). Neurological conditions in this group matched those in the BP+ND group.

Controls. Twenty-two normal individuals with serum samples also collected from 2001 to 2008.

Informed consent was obtained from each patient for blood sampling. The research was in compliance with the Declaration of Helsinki. The procedures to obtain human skin and brain tissues were approved by the Ethics Committee of PUMCH (Protocol S-300). Informed consents were obtained from these donors.

Serum Collection

Venous blood was collected from patients at disease diagnosis (BP and/or ND). Serum was prepared and stored at –70°C until use.

Protein Extract Preparation

Foreskin samples were obtained from healthy individuals. The epidermis was separated from the dermis by incubating the skin with 1 M NaCl, containing 1 mM phenylmethylsulfonyl fluoride (PMSF) and 1 mM ethylenediaminetetraacetate at 4°C for 72 h. Normal human brain tissue was obtained from donors undergoing temporal lobe resection for medication-resistant epilepsy treatment. Both epidermis and brain tissues were homogenized in 6-part sample buffer [65 mM Tris-Cl (pH 6.8), 65 mM Tris-Cl (pH 8.0), 0.5% SDS, 0.1% NP-40, 2 mM PMSF and Calbiochem Protease Inhibitor Cocktail Set III (Merck, Shanghai, China)]. The homogenates were ultrasonicated (4 × 15 s), and centrifuged at 12,000 *g* for 15 min at 4°C. Protein concentrations of the supernatants were then determined by Bradford assay using BSA as standard protein. Supernatants were stored at –70°C until use.

SDS-Polyacrylamide Gel Electrophoresis and Immunoblotting

Total protein extracts (80 μg) were mixed with equal volumes of loading buffer [12 mM Tris-HCl (pH 6.8), 10% SDS, 20% glycerol, 10% β-mercaptoethanol, and 0.01% bromophenol blue], heated at 100°C for 3 min, and separated by 5% SDS-polyacrylamide gel electrophoresis (SDS-PAGE). Proteins were then transferred to nitrocellulose membranes with a trans-blot cell apparatus (Bio-Rad Laboratories, Beijing, China) at 45 V overnight. Transfer efficiency was verified by 0.2% Ponceau S red (Sigma-Aldrich, Shanghai, China) staining. In order to test reactivities of protein extracts with various serum samples of patients, blotted nitrocellulose membranes were cut into strips and blocked for 1 h at room temperature in blocking buffer [5% skimmed milk in TBS-T (10 mM Tris-HCl, 0.3% Tween-20)]. Subsequently, blots were either incubated with a positive control anti-BPAG1 antibody (1:300), BPAG1n3 (Santa Cruz Biotechnology, Santa Cruz, Calif., USA), or with serum samples diluted at 1:50 in blocking buffer. Incubation was carried out at room temperature for 1 h with gentle agitation. Blots were then washed in TBS-T (3 × 10 min) before incubating with horseradish peroxidase-conjugated

goat anti-human IgG (Santa Cruz Biotechnology) diluted at 1:2,500 in blocking buffer at room temperature for 1 h. Blots were then washed in TBS-T (3 × 10 min) and subjected to enhanced chemiluminescence assay (Santa Cruz Biotechnology). Each serum sample was tested at least twice.

Statistical Analyses

ANOVA (positive rate) and χ^2 tests were used for the overall comparison among groups, and Bonferroni correction was used for the multiple comparison; $p \leq 0.05$ was considered statistically significant. All statistical analyses were performed with the SAS software version 6.12 (SAS Institute Inc, Cary, N.C., USA).

Results

Late Onset of BP Relative to ND

Analysis of the patients' clinical data revealed that the diagnoses of BP in all of these 22 elderly BP+ND patients were at the mean age of 77.7 ± 7.9 (60–97) years, whereas the average age of onset of ND was at 72.0 ± 7.3 (55–88) years (table 1). The onset of BP was after ND in all of these patients. The average interval between ND and BP was 5.7 ± 2.6 (2–12) years (table 1).

Epidermal and Brain Extracts Contain BP Antigens That Could Be Recognized by Serum Samples of BP Patients

In order to examine if BP antigens exist in human skin and brain extract, and can be recognized by sera of BP patients, blots containing human epidermal and brain extracts were incubated with serum samples of 22 BP patients and 22 normal control individuals. Polyclonal antibody BPAG1n3, which specifically recognizes the neural isoforms of BPAG1, was used as the control antibody. Serum samples of a proportion of BP patients could recognize a 230-kDa band in the epidermal (50.0%) and brain extracts (9.1%) (fig. 1). Representative positive blots are shown in figure 1. The BPAG1n3 antibody could also recognize the 230-kDa band in both epidermal (fig. 1a) and brain extracts (fig. 1b), whereas none of the serum samples of the controls resulted in any positive signals with the epidermal extract (fig. 1a, 2a) and merely 4.5% with the brain extracts (fig. 2a). A 180- and a 165-kDa protein were also recognized by BP patients' sera; 59.1% of BP patients' serum samples reacted with these proteins in the epidermis and 18.2% in the brain (data not shown).

BP+ND Patients' Serum Samples Could Recognize BP Antigens in Human Epidermal and Brain Extracts

In order to examine if BP+ND patients' sera contain autoantibodies that could readily react with BP antigens in the epidermis and the brain, serum samples of these patients were incubated with blots containing human epidermis and brain protein extracts. A significantly high proportion (72.7 and 54.5%, respectively) of BP+ND patients' sera could recognize the antigens in the epidermal and brain extracts (fig. 2a). Specific to the brain, 12 out of 22 BP+ND serum samples showed positive bands at 230 kDa (fig. 3). In contrast, none of ND patients' sera reacted with the epidermal extract, and a substantially low number with the brain extracts (0 and 9.1%, respectively) (fig. 2a). BP+ND patients' sera could also recognize the 180- and 165-kDa antigens in the epidermal and brain extracts (63.6 and 59.1%, respectively; data not shown).

The numbers of serum samples that reacted positively with human brain extracts were compared among the BP+ND, BP, ND and control groups. We found that significant differences exist amongst these 4 groups (χ^2 analysis; $p = 6.243 \times 10^{-5}$). When BP+ND and BP,

Table 1. Clinical information of BP+ND patients

Patient No.	Sex	Age at BP onset years	Age at ND onset years	BP and ND interval years	Type of ND
1	M	97	88	9	CD (VD)
2	F	79	71	8	CD (VD)
3	M	84	74	10	CD (VD)
4	M	76	70	6	CD (VD)
5	F	84	79	5	CD (VD)
6	M	76	70	6	CD (VD)
7	M	81	77	4	AD
8	M	76	69	7	AD
9	M	72	70	2	CD (CH)
10	M	68	65	3	CD (CH)
11	M	82	70	12	CD (CH)
12	F	66	60	6	CD (CH)
13	M	67	63	4	CD (CH)
14	M	85	82	3	CD (CT)
15	M	78	73	5	CD (CT)
16	M	84	81	3	CD (CT)
17	M	81	71	10	CD (CT)
18	F	75	71	4	CD (CT)
19	M	81	75	6	CD (CT)
20	M	60	55	5	CD (CT)
21	M	79	75	4	CD (CT)
22	M	78	74	4	CD (CT)
Mean	..	77.7 ± 7.9	72.0 ± 7.3	5.7 ± 2.6	..

CD = Cerebrovascular disease; VD = vascular dementia; CH = cerebral hemorrhage; CT = cerebral thrombosis; AD = Alzheimer's disease.

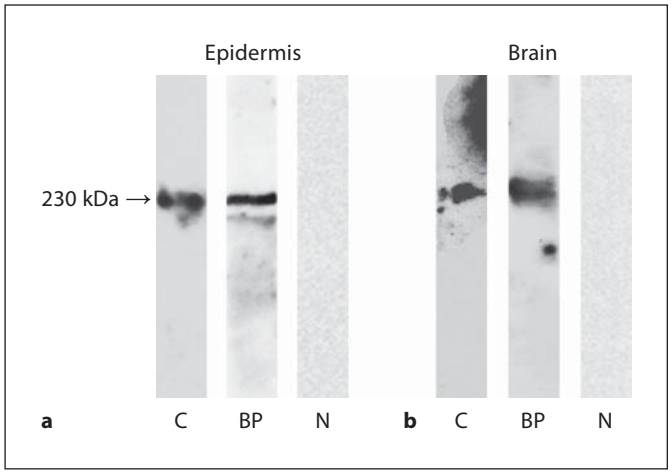


Fig. 1. Sera of BP patients readily react with BPAG1 (BP230) in human epidermis and brain protein extracts. **a** A 230-kDa protein in the human epidermis was recognized by the positive control antibody (C), BPAG1n3, and 50% of the serum samples of BP patients, but not normal control individuals (N). A representative positive 230-kDa band is shown for the BP group. **b** A 230-kDa protein in the human brain was recognized by the positive control antibody (as in **a**), 9.0% of the serum samples of BP patients, and 4.5% of controls. Representative serum samples reacting positively with BP230 are shown for the BP group.

BP+ND and ND, or BP+ND and control groups were compared separately, there were also significant differences ($p = 0.0012$, 0.0012, and 0.0003, respectively; fig. 2b). However, there was no significant difference between the BP and ND, BP and normal, or ND and normal groups (fig. 2b). In the epidermis, no significant difference ($p = 0.1216$) existed between BP+ND and BP patients (fig. 2b).

Discussion

The amino acid sequence of BPAG1 is highly (96%) conserved between mouse and human [8]. In a previous study, we found that the 230- and 180-kDa proteins of the mouse brain can be recognized by the serum samples of BP+ND patients [6]. In the present study, we observed that serum samples of BP+ND patients could also recognize the corresponding proteins in the human brain. Therefore, the current study further proved that BP antibodies in this group of patients could cross-react with antigens in both skin and brain.

We first attempted to use immunohistochemistry assays to examine whether serum samples of BP+ND pa-

Substrate	Epidermis				Brain			
Patient	BP+ND	BP	ND	N	BP+ND	BP	ND	N
230 kDa positive (/total)	16/22	11/22	0/22	0/22	12/22	2/22	2/22	1/22
Percentage	72.7	50.0	0.0	0.0	54.5	9.1	9.1	4.5

a

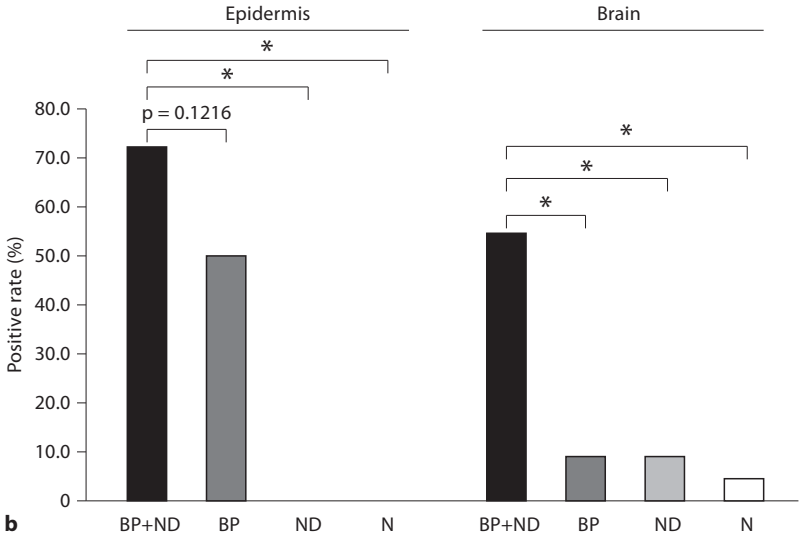


Fig. 2. The number of serum samples of all 4 groups of patients that could recognize BPAG1 (BP230) in human epidermis and brain extracts. **a** The number and percentage of serum samples that positively reacted with the 230-kDa protein (BPAG1). **b** Percentage of serum samples that positively reacted with the BPAG1. * $p < 0.01$.

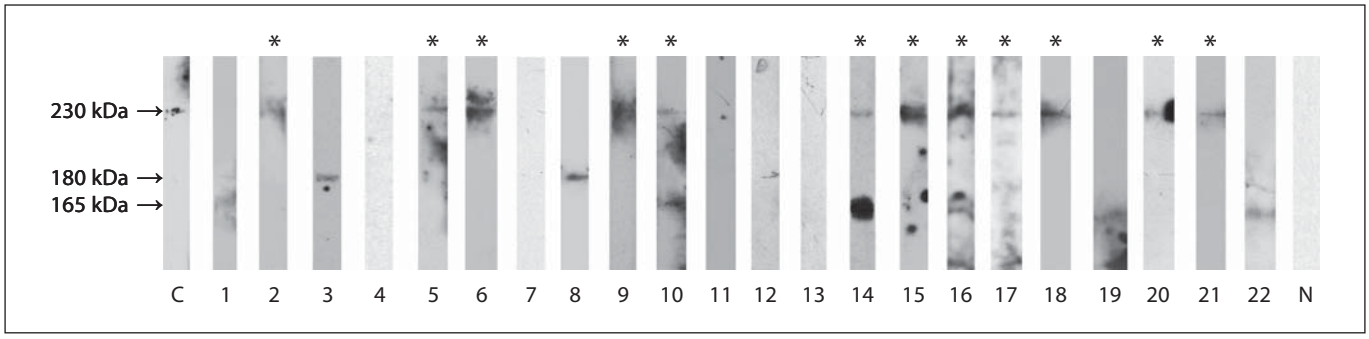


Fig. 3. Immunoblots of the human brain extract incubated with serum samples of all 22 BP+ND patients. 12 out of 22 (1–22) samples (54.5%) reacted with a 230-kDa protein (*). A proportion of 180- or 165-kDa proteins was also recognized by these serum samples. C = Control antibody; N = normal controls.

tients could react with antigens in human brain tissue. However, we were not able to obtain any positive signals. Alternatively, we utilized the immunoblotting technique as described in the 'Methods' section. We observed that 54.5% of the serum samples of BP+ND patients could react with human brain extracts, which is in contrast to only 9.1% of BP patients. This result suggested that the sera of BP+ND patients contain antibodies that readily react with antigens in the brain, and that antibodies against the neural form of BPAG1 could cross-react with that of the epidermal form.

Neurological disorders associated with the BP patients are often neurodegenerative diseases [1, 3]. There is evidence indicating the existence of anti-brain autoantibodies in the serum of Alzheimer's disease and cerebral stroke patients [9]. MRI has revealed that during the acute phase of neurological disorders such as multiple sclerosis, damage can happen to the blood-brain barrier as a result of multiple microtraumas, microvascular pathology, or local inflammation [10]. A compromised blood-brain barrier may result in the abolishment of the 'immune privilege' of the brain. Therefore, we speculate that neurological conditions in BP+ND patients could result in the exposure of brain antigens, and consequently the generation of autoantibodies such as BPAG1. These autoantibodies may or may not further contribute to the underlying ND. However, over time, they may be amplified, resulting in skin manifestations of BP. This may also explain the fact that in BP+ND patients, the diagnosis of BP always follows that of ND [1].

In the central nervous system, BPAG1 and BPAG2 are colocalized at the synaptic and extrasynaptic regions [11, 12]. Both BPAG1 and BPAG2 are considered components of the neural-specific anchoring complex to which central nervous system laminins bind to stabilize presynaptic terminals to the underlying extracellular matrix. This composition resembles the laminin-anchoring complex of the skin, in which BPAG1 and BPAG2 play critical roles in stabilizing the dermal-epidermal junction [11, 12]. BPAG1- and BPAG2-associated defects in the laminin-anchoring complex in the synapse may contribute to the neurological symptoms observed in BP+ND patients [11–13].

In a previous study, we found that some BP+ND patients' sera recognized 180- and 165-kDa proteins from mouse brain extracts [6]. In the present study, we observed that serum samples of BP+ND patients could also react with antigens of 180 and 165 kDa in the brain extract, suggesting the existence of BPAG2.

In conclusion, our data indicated that antibodies in BP+ND patients' sera could recognize BPAG1 in both the human epidermis and brain. This observation raised the possibility that BPAG1 could act as a shared autoantigen in both ND and BP. It is possible that alterations in the central nervous system in the course of ND could expose the neuronal isoform of BPAG1, triggering an immune reaction that, along with the immunological cross-reactions, results in the development of BP. Understanding the link between these two diseases will undoubtedly contribute to the development of preventive and treatment strategies for elderly BP patients.

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